Stop Heating an Empty Building: Reward Anatomy and a Supervisory Layer for the NeurIPS 2026 Smart Buildings Challenge

Semyon Morev
Institute of Computer Science
NUST MISIS, Moscow, Russia
m2508348@edu.misis.ru

Abstract

The NeurIPS 2026 Smart Buildings Challenge asks for one HVAC controller that runs, without retraining, in commercial buildings it has never seen. We show that three properties of the challenge's reward decide what such a controller should do. Comfort is charged as a mean over the occupants present, so being ready when people arrive matters most and an empty building need not be heated. A mandatory share of outdoor air and a capped supply temperature make heat cheaper to deliver into a colder building, in one burst. And every zone holds the same number of people, while the heat it needs grows with its floor area. Starting from Wu's public adaptive-physics controller, we add a supervisory layer in three steps: a readiness ramp with a release on days nobody comes; no heating of the empty building at night, above a frost floor; and thirteen smaller mechanisms found by reading simulation traces, among them comfort margins scaled by zone size. The result is $5.9\,\%$ better than the public controller on the simulator with its billing errors corrected, and the thirteen mechanisms improve every one of 49 held-out scenarios; on the public leaderboard the score moves from 0.01062 to 0.01002. We also measure what missing information would be worth—a per-zone occupancy sensor about $3\,\%$, a perfect weather forecast nothing—and audit the simulator's billing errors, which reward an exploit and overcharge an honest controller by about $2\,\%$.

1 Introduction

The Smart Buildings Challenge [9] evaluates controllers under a protocol that makes transfer the whole problem: one policy is instantiated afresh for each of 30 scenarios—six buildings crossed with five weather stations—and must run 8,640 steps of five simulated minutes without retraining, network access or a site visit. Zone count varies threefold ($V \in [30, 85]$) and air-handler count threefold ($H \in [3, 9]$), so the action dimension, $3H + 2 + 2V$ continuous values plus $H + 1$ binaries, changes per scenario; no physics, plant sizes or per-zone occupancy are exposed; and a submission gets one 9-hour session for all 30 scenarios [16], about 125 ms per step. End-to-end reinforcement learning is therefore expensive (about 107 CPU-hours for 5×10^6 PPO steps), and the organizers report only single-digit-percent gains from SAC on one building [7]. Meanwhile the leaderboard is public, so the object to beat is a documented notebook: the strongest we found [26] scores 0.0075040 on the 30 shipped scenarios (0.01062 on the leaderboard) against 0.05498 for the organizers' `pid_water`. We port that controller bit-exactly and ask what is left to fix once a strong public controller is in hand.

Three properties of the reward. Comfort is charged to *present* occupants only, and as a mean, so the first arrivals cost as much per kelvin as the peak: readiness at the mode change is a first-order

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objective and the night is free. The bill is envelope heat times a multiplier that the $30\,\%$ outside-air floor and the 308.15 K discharge cap fix (Eq. 2)—so the same recovered heat is cheaper when it is delivered into a colder building, in one burst, at the coil's cap. The consequence is a two-number change to a public controller. CtrlW adds a 2 h readiness ramp, a release of the building nobody came to and reward-consistent fuel weights; CtrlW5 then stops heating the empty building, because the inherited 282 K standby floor, held by the coldest zone of each air-handler group, keeps every group's plant on from 19:00 to 04:00. Releasing it plus a cooler hot-water loop is worth 12.8 of the 93.1 units lost on the seven-day *anchor* (§2), and 6.4 with the 278 K ($4.9\,^{\circ}$C) floor we ship. A receding-horizon planner we built to look for the same gain does not find it, which is why we present this as a *supervisory sequencing* result rather than a control-quality one. CtrlX continues in the same way: under a fixed benchmark on the corrected simulator, two evolutionary searches over its knobs found nothing and thirteen mechanisms read from traces found another $2.9\,\%$, the last of them from a third fact about the reward—every zone holds the same headcount, so a kelvin of comfort is worth the same in a small zone and a large one, and costs heat in proportion to floor area.

Contributions. (1) A reward anatomy: comfort is a per-occupant mean, and the bill is envelope heat times a closed-form ventilation multiplier (§3). (2) A supervisory layer on the public controller in three steps (CtrlW, CtrlW5, CtrlX), with ablations, a frost-floor cost curve and the refutations that bound it (§4). (3) An evaluation on the 30 training scenarios, 49 held-out ones, both simulator versions, reseeded occupancy and the leaderboard (§5). (4) A measured value of the missing information and an audit of the simulator's billing in both directions (§6, §7). Code, per-scenario results and the study reports are public.[1]

2 The benchmark and related work

A scenario is a partially observed Markov decision process over a temperature field, plant parameters and hidden per-zone occupancy; the policy acts in $\mathcal{A} = \mathbb{R}^{3H+2+2V} \times \{0,1\}^{H+1}$ (per handler a supply-air temperature, a static-pressure setpoint and a recirculation fraction $r_k \le 0.7$; two plant values; per zone a damper floored at $10\,\%$ and a reheat valve), observes $6V + 15H + 58$ values with zone readings lagged one step and *one* building-level occupant count, but no per-zone occupancy and no weather forecast; the plant cools centrally and heats only through per-zone reheat coils. With live setpoints $T_v^{\mathrm{lo}}, T_v^{\mathrm{hi}}$ and per-zone shortfall $s_{v,t}$ (linear to $w = 5$ K, quadratic beyond), the step reward and score are $r_t = -0.6\,C_t - 0.2\,\hat{E}_t - 0.2\,\hat{C}_t \le 0$ and $S = \frac{1}{30}\sum_{s=1}^{30} \left|\frac{1}{T}\sum_{t=1}^{T} r_{s,t}\right|$, with C_t the occupancy-weighted shortfall of Eq. (1) and $\hat{E}_t, \hat{C}_t$ metered electricity-plus-gas cost and carbon normalized by the draw at rated power times H; lower S is better. One runaway zone therefore dominates an episode, and the denominators scale with H, so one parameter set behaves consistently across building sizes.

Terms used throughout. `dev30` is the 30 training scenarios at full 30-day length; the *holdouts* are 49 scenarios on unseen generated buildings and unseen climates (§5). The *anchor* is the seven-day `fairbanks_1` episode on which the public forum compares controllers, reported as a total reward in *units* (Appendix A). The *shipped* simulator is the starter kit as released; the *corrected* one adds two open pull requests that fix how it bills boiler gas (B1) and mixes return air (B2), the two accounting *seams* audited in §7.

The challenge [9] asks whether domain-specific models or generic foundation models adapt better to unseen buildings, on the substrate of the Smart Buildings Control Suite [7] with its lightweight finite-difference simulator [10]; its authors concluded that aggregating a policy into a static schedule retains most of its value [6] and that better simple heuristics remain to be found [8], which is the line our result continues. Because the leaderboard is public we measure against three notebooks rather than re-derive them—a per-zone PI rule [13], a residual PPO policy on it [4], and the analytic one-step search we adopt [26]—all rule-based or one step deep. Transfer is formalized by Building2Building [24] and attacked by GenTL [20] and BUILDA [12]; in practice MPC is limited by the modelling it requires [5], so we keep the base controller's fixed per-zone model, online residual and one-step search, which stands in for the reset sequences of Guideline 36 [2] and trim-and-respond [25]. Evolution strategies suit a slow deterministic simulator [11, 21]; CityLearn 2022 [17] is the closest precedent, and benchmark hygiene follows [3, 15].

[1] `https://github.com/MorS1337/sb26-stop-heating`

3 Reward anatomy

Comfort regret is a per-occupant mean, so arrival dominates. For zone v with live band $[T_v^{\text{lo}}, T_v^{\text{hi}}]$ and shortfall $s_{v,t}$ (linear to 5 K, quadratic beyond), the productivity term of the reward is

$$\frac{P - P^{\min}}{P^{\max} - P^{\min}} - 1 \;=\; -\frac{\sum_v o_{v,t}\, s_{v,t}}{\sum_v o_{v,t}}, \tag{1}$$

the occupancy-weighted *mean* shortfall over the occupants present [9]. The headcount cancels: a kelvin of shortfall costs the same with two people in the building as with sixty. Occupancy is exogenous—the same headcount in every zone (two in most scenarios, one in the rest), arrivals 06:00–12:00, departures 13:00–19:00—so the ramp runs from 0 to $2V$ people across the morning (74 on the anchor) and a reactive controller pays a fixed price at every mode change. The measurement is one horizon change: the public base controller's preheat gain is multiplied by a parameter that lives inside a 60 min look-ahead flag, so its ramps never exceeded one hour. Moving the ramp outside the flag at a 2 h horizon cuts mean hour-06 comfort regret from 46.2×10^{-4} to 9.2×10^{-4} per step and the score from 0.0075040 to 0.0074263; the remaining $0.0074263 \to 0.0073694$ is the release of a building that is still empty at noon, which is additive with it ($-0.77\,\%$ alone). Longer horizons keep improving the arrival hour while the score rises (0.0074906 at 3 h, 0.0075614 at 4 h): readiness saturates once every zone is in band; the gas does not.

The bill is envelope heat times a ventilation multiplier. On the public forum's comparison episode (`building_train_1/fairbanks_1`, 2 016 steps, mean outside $-26.3\,^\circ$C) the same controller loses 93.12 units: comfort 0.54 (0.6 %), energy 26.23 (28.2 %), carbon 66.35 (71.3 %), carbon being 2.5 times energy. Instrumenting the trajectory at the model call partitions the bill: $73.3\,\%$ envelope reheat at the band bottom, $22.9\,\%$ the outside-air share of the same reheat, $1.9\,\%$ fans, $1.3\,\%$ boiler standing loss, $0.6\,\%$ pump, no chiller. That outside-air share is not a modelling choice but the action space's floor: recirculation is capped at $r \le 0.7$, so at least $30\,\%$ of the supply air is outdoor air, and the coil must lift it from $T_{\text{mix}} = 0.7\,T_{\text{ret}} + 0.3\,T_{\text{out}}$ to the discharge cap $T_{\text{dis}} = 308.15$ K while the zone only receives the drop from T_{dis} to T_{zone}. With $T_{\text{ret}} \approx T_{\text{zone}}$ the plant's own air-side law is

$$\frac{\text{gas billed}}{\text{net zone heat}} \;=\; 1 + \frac{0.3\,\big(T_{\text{zone}} - T_{\text{out}}\big)}{T_{\text{dis}} - T_{\text{zone}}}, \tag{2}$$

so at the anchor's ambient the multiplier is 2.00 for a zone held at 294 K and 1.18 for one at 270 K (Figure 1a). *Heat delivered into a colder building is cheaper per delivered watt.* What a controller cannot change is the 0.3; what it can is the temperature at which it delivers the recovery. Measured over the episode, gas per net zone watt is 1.975 for CtrlW and 2.061 for CtrlW5 against a cap-limited 1.949 and 2.045: the air side has about one unit of slack (1.15 units, 1.2 %, for CtrlW; 0.58 for CtrlW5), so *airflow is not where the money is*. The ventilation share is large in the counterfactual sense (coil heat against non-ventilation load, a factor 2.24), but that only describes how the bill splits; no controller can move it.

What the anatomy rules out. The exchange rate between the two terms forbids trading the occupied band: on the anchor 624 of the 2,016 steps have at least one occupant, and a uniform 1 K shortfall there costs $0.6 \times (1/5) \times 624 \approx 75$ reward units, while one kelvin of building temperature is 0.57 MWh of gas $\approx$ 1.9 units—an $\approx$ 39:1 ratio (measured: holding the schedule's own setpoint instead of 0.12 K above it costs 1.53 units). What is left is the band nobody occupies: the night is free—and CtrlW, like the family it comes from, heats it anyway.

4 Method: from a public controller to a night release

4.1 The base controller

We adopt the public controller of Wu [26], credit it as such, and reproduce it exactly: with all our flags off, our port returns its dev30 score to the last digit (0.0075040). It gives each zone one first-order model whose coefficients are set from the zone's geometry and never updated, plus a residual that tracks the one-step error, lagged one step to match the sensor delay; evaluates candidate (supply-air temperature, recirculation) pairs analytically against those models per handler, a one-step-deep MPC; corrects drift with a small MLP ensemble; and weights predicted gas 0.30 against electricity.

It is strong (comfort 2.14×10^{-4} per step on dev30) but plans neither the morning nor the night: its standby policy holds each zone between 282 and 307 K, and because a group's plant runs when *any* zone of the group calls, one cold zone holds every group's plant on all night.

4.2 CtrlW: readiness, weights, guards

Readiness ramp. The base controller ramps its band from the eco toward the comfort window over `precondition_hours`, but the ramp lives inside the code path guarded by `comfort_mode_soon`—a flag true only in the 60 minutes before comfort-mode onset—so any horizon longer than 1 h was clamped by construction. We run the same ramp outside the flag: from h before the mode change t_s, each zone's lower bound is $T^{\text{lo}}_{\text{eco}} + p(t)\,(T^{\text{lo}} - T^{\text{lo}}_{\text{eco}}) + \mu$ with $p(t) = 1 - (t_s - t)/h$ and μ the comfort margin (the upper bound alike), each step's request capped by the base controller's recovery clamp at $1/1.5$ of the remaining gap; the one-step search tracks this band, and $h = 2$ h is the tuned horizon.

Weights and guards. Per unit of gas energy the reward charges $0.2\,(p_g + \lambda\kappa_g)$ and per unit of electricity $0.2\,(p_e + \lambda\kappa_e)$ (p price, κ carbon intensity, λ the ratio of the cost to the carbon normalizer, up to a common scale), so the consistent relative weight is $(p_g + \lambda\kappa_g)/(p_e + \lambda\kappa_e) \in [0.443, 0.509]$ on the 30 shipped scenarios, mean 0.481: the base under-weights gas by about $38\,\%$. Measured, it is a plateau (0.0075035, better in 16 of 30) because gas is spent on reheat during warm-up and electricity on fans and chillers during the day; we adopt the derived weight because it is the correct objective and claim no gain for it. Of the two guards, the occupancy-aware release fires when the building count is still zero at noon, which happens on public holidays: 0.0074459 alone ($-0.77\,\%$, better on 23 scenarios, identical on 7), and §4.5 moves it to the morning. The holiday guard is inert: the schedule's own holiday set is empty.

What tuning the class buys. A per-zone horizon (*CtrlW2*: 0.0073576, $-0.16\,\%$, better on 12, worse on 6) and a fixed 2 K pre-dawn lift for zones on the standby floor (*CtrlW4*: 0.0073445, $-0.34\,\%$, better on 8, worse on 10) are both small.[2]

4.3 CtrlW5: do not heat an empty building

The night is unoccupied by construction (arrivals 06:00–12:00), so comfort charges nothing there, yet CtrlW holds each group's coldest zone at 282 K from 19:00 to 04:00. CtrlW5 changes exactly two numbers in `PARAMS`: `standby_minimum` $282 \to 262$ K, which releases the floor, and `hot_water_target` $345 \to 336.32$ K. Restoring both reproduces CtrlW4 bit-exactly, and the two are worth 12.07 and 0.75 units of the 12.82: the single floor release is the mechanism, the cooler loop is a bonus. Three mechanisms, all measured on the anchor (Table 2, Figure 1): *(a) the night heating stops*—night coil duty falls from 30 kW to 0 kW while the building's *mean* temperature barely moves ($287.9 \to 287.4$ K, i.e. $14\,^\circ$C, still 2 K from the schedule's own 289 K night setpoint: it is the coldest zones and the group plant that change, not the building); *(b) the recovery is one burst at the discharge cap*—by Eq. (2) heat delivered into a colder mass is cheaper per delivered watt, and CtrlW5's discharge sits at $35.00\,^\circ$C with 1.1 K mean headroom against CtrlW's 2.3 K; *(c) the stored heat pays for the day*—charging the mass at 04:00–06:00, the cheapest hours (88 kg CO_2/MWh), and letting it drift through the band costs nothing in comfort, because the band's upper edge is free, and removes much of the continuous daytime coil duty.

Net: gas $27.19 \to 23.27$ MWh ($-14.4\,\%$), comfort $-0.54 \to -1.12$; the $+12.82$ units are envelope reheat $+8.75$, the outside-air share $+4.42$, fans $+0.27$, pump -0.22, boiler standing loss $+0.18$ and comfort -0.58 (by hour block in Figure 1b). A search over the same family and a floor sweep agree that this is the family's optimum rather than a lucky knob, though neither rules out a better mechanism outside it. A 150-evaluation CMA-ES over CtrlW4's 16 supervisory knobs reaches -83.109 with its optimum on the lower bound of `standby_minimum`, and releasing that bound takes it to -80.298, i.e. to CtrlW5-262's -80.306 within 0.008 units. A floor sweep at that optimum's other knobs is monotone and saturates at 262 K, where the floor stops binding and the building free-floats: the optimum is *no overnight heating at all*, and the comfort term stays at -1.35 from 274 K down (CtrlW5 keeps CtrlW4's other knobs and ends at -1.12), so the floor's gain is not bought from the occupied band.

[2] CtrlW3, CtrlW6 and CtrlW7 are refuted variants, reported in §5, Appendix G and §4.5.

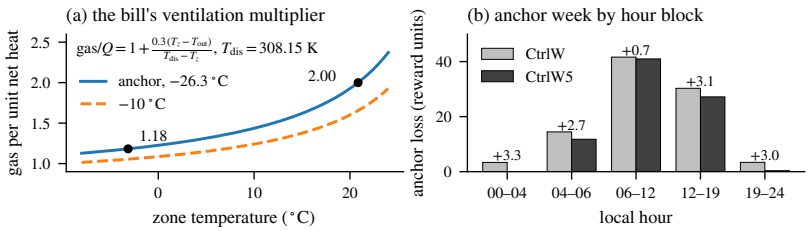


Figure 1: **(a)** The bill's ventilation multiplier, Eq. (2), at the anchor's ambient ($-26.3\,^{\circ}$C) and at $-10\,^{\circ}$C: the same delivered heat costs 2.00 units of gas at a 294 K zone and 1.18 at 270 K, which is why releasing the night and recovering in one burst pays. **(b)** The anchor's loss by hour block: releasing the floor removes 3.3 (00–04), 2.7 (04–06) and 3.0 (19–24) units from the blocks the plant used to heat, and 0.7 and 3.1 from the two daytime blocks, where the stored charge replaces coil duty. Bar heights are losses (lower is better) and the labels are savings.

4.4 Frost protection, and what it costs

A released building in Fairbanks does not stay above freezing on its own: the 262 K release gains 12.8 units by letting the coldest room reach $-8.58\,^{\circ}$C on the anchor and $-11.2\,^{\circ}$C on a 30-day episode, with 19 and 90 hours in which some room is below $0\,^{\circ}$C. The simulator has no freezing model and the score has no penalty for it, so protecting the building is our choice. We ship a floor: **278 K**, i.e. a $4.85\,^{\circ}$C setpoint floor, which keeps the coldest room at $+4.66\,^{\circ}$C on the anchor and $+4.56\,^{\circ}$C on the three cold 30-day scenarios, with zero room-hours below $0\,^{\circ}$C, and still beats CtrlW by 6.9 % on the anchor and 3.8 % on the six Fairbanks scenarios (Appendix H). This is a measured minimum on four scenarios, not a guarantee: the floor is a setpoint the controller can miss. The price is steep at the top—282→280 K buys 3.5 units, 280→278 another 2.9—and flat at the bottom (Figure 3, Appendix H); $\geq 5\,^{\circ}$C needs 280 K and costs 9.3 of the 12.8 units, and 274 K leaves rooms at $+0.63\,^{\circ}$C.

4.5 CtrlX: thirteen mechanisms read from traces

CtrlW5-278 then went under a fixed protocol: a ten-scenario, seven-day benchmark on the corrected tree with a non-default occupancy seed; a module that stays numpy-only, reads no building or weather name, keeps every room above $+4\,^{\circ}$C and touches neither accounting seam; and paired 30-day runs for everything kept. Two CMA-ES campaigns over 14 knobs found nothing; every gain came from reading a trace until it showed a mechanism. Table 4 (Appendix C) lists the thirteen with a leave-one-out ablation; three carry §3 further.

(i) Holiday release. The schedule runs comfort mode on every weekday, but the occupancy model keeps people home on public holidays (the building is still empty at noon in 23 of the 30 `dev30` windows), and the inherited release waits for noon. Arrivals are about $1/36$ per occupant per step, so a building still empty $K = \lceil 330/V \rceil$ steps after the mode change is empty for the day with a false-release probability below 10^{-4}. Releasing then is the largest mechanism (1.0 %).

(ii) A late start when the fabric is still warm. The zone air is light and the fabric slow: 3 kW moves one zone's air 2.5 K in a step, the air falls 1.5 K in the first step after the plant stops, and the fabric creeps 0.07 K per step. A pre-arrival ramp is therefore a fabric charge at a low air temperature—cheap in deep cold by Eq. (2)—but every step of it ventilates every zone of the group at the floor flow. Both sides scale with $T_{\text{zone}} - T_{\text{out}}$, so the break-even is set by the fabric's charge alone: to first order a gap of 2–5 K between the group's mean temperature and the setpoint (Appendix C). A group within 6 K when the window opens drops the 2–3 h ramps and keeps the base controller's last hour. The same picture explains two refutations: the frost floor cannot be duty-cycled (the air falls to a fabric that sits below the floor: four designs cost +25 to +300 % of a cold Saturday, and an earlier latch 0.3–4.3 anchor units with rooms dipping to $+1.1\,^{\circ}$C), and asking the one-step inversion for more than its clamped step to arrive on time loses 0.4–53 %.

(iii) Margins scaled by zone size. Eq. (1) weighs a zone by its occupants and every zone holds the same number, so a kelvin of shortfall costs the same in a 40 m^2 zone and a 360 m^2 one, while the heat that prevents it scales with floor area, which spans 3.5–9× inside one building. The comfort margin

Table 1: Main comparison; score S (lower is better). `dev30` is 30 shipped scenarios at 8,641 steps, `gen` 32 held-out generated scenarios and `ext` 17 unseen climates (shipped tree); *shipped/corrected* are the two simulator trees. CtrlX is the controller we ship; both it and CtrlW5-278 keep a 278 K night floor, CtrlW5-262 does not. The tank hold responds to the shipped accounting and is reported apart (§7). Public rows are the authors' leaderboard scores unless we re-ran the controller.

Method	dev30 shipped	dev30 corrected	gen / ext	Kaggle public
`pid_water`	0.05498	0.05789	0.03620 / 0.05189	—
per-zone PI [13]	0.01042	0.01023	0.007673 / 0.010748	0.01420
Wu [26]	0.0075040	0.0072641	0.005293 / 0.008015	0.01062
CtrlW	0.0073694	0.0071449	0.005179 / 0.007836	0.01051
CtrlW4	0.0073445	0.0071587	0.005170 / 0.007813	0.01049
CtrlW5-278	0.0071980	0.0070409	0.005058 / 0.007616	0.01032
CtrlW5-262 (no frost floor)	0.0071338	0.0069430	0.005052 / 0.007577	0.01024
CtrlX	**0.0069712**	**0.0068387**	**0.004840 / 0.007353**	**0.01002**
CtrlX + tank hold	0.0068269	0.0068420	0.004713 / 0.007173	0.00982

and the arrival lead, uniform in kelvin and minutes until then, are scaled per zone by $(\bar{A}/A_v)^{1/2}$ (median area over the zone's own, clipped to $[\frac{1}{3}, 3]$). Comfort, energy and carbon improve together on the benchmark, where every uniform change of margin or lead had traded them one for one; the exponent is a plateau (0.25–0.5 within $0.01\,\%$, 1.0 over-reaches).

5 Results

Protocol. `dev30` (8,641 steps per scenario) is what the leaderboard draws from; `gen` is 32 scenarios from eight generated buildings under weather windows absent from tuning; `ext` replays those buildings and two shipped ones under 17 Meteostat 2023 stations absent from the benchmark (Appendix J). Scores are means over scenarios of $|\frac{1}{T}\sum_t r_t|$ and comparisons against a base are paired. Reseeding occupancy moves a score by at most $0.18\,\%$ and a paired gain by at most 0.12 percentage points (below), so we read only larger deltas as gains. Numbers are shipped unless marked corrected; we report both, because the organizers aim to run final verification on the corrected simulator [19, 22, 23] with a new occupancy seed [18], and neither seam nor seed is something we tune against. The CtrlX runs use an exact fused C step of the simulator—observations and rewards bit-identical on twelve gate episodes and full-length traces, 1.6–$2.1\times$ faster (Appendix F).

Main comparison. Table 1. On dev30 the organizer baseline scores 0.05498, the per-zone PI rule 0.01042, the base controller 0.0075040, CtrlW 0.0073694 and CtrlW5-278 0.0071980; the corrected tree keeps the order (bar CtrlW4, now just behind CtrlW) and moves the non-organizer controllers by -1.8 to $-3.2\,\%$ (the tank hold apart) and makes `pid_water` worse. On the holdouts CtrlW5-278 is $2.6\,\%$ better than CtrlW—better on 38 of 49 scenarios, identical on 11, worse on none (sign test $p < 10^{-11}$)—so the gain is not a dev30 artefact; the unseen climates are $51\,\%$ worse than the same buildings under in-distribution weather (0.007616 against 0.005058), so the transfer gap comes from the climate. On the public leaderboard CtrlW scored 0.01051 (submission 58025908), CtrlW4 0.01049, CtrlW5-278 0.01032 (58059955) and CtrlW5-262 0.01024 (58050174) against the base controller's 0.01062; the leaderboard/offline ratio is 1.36–1.44 across the eight methods with both numbers, which we use only as a consistency check. The frost floor costs $0.8\,\%$ on the leaderboard, as it costs $0.9\,\%$ on dev30.

CtrlX. On the corrected tree, where it was developed, CtrlX scores 0.0068387 on dev30 against CtrlW5-278's 0.0070409 ($-2.87\,\%$; better on 29 of 30, the other $+0.04\,\%$) and 0.0055822 against 0.0057936 on the 49 holdouts ($-3.65\,\%$; better on all 49, from $-0.9\,\%$ in the hot months to $-11.8\,\%$ in the shoulder ones), so the gain is larger away from the development set than on it. The two mechanisms added last transfer one to one: the off-peak tank floor is $-0.033\,\%$ on dev30 and $-0.036\,\%$ on the holdouts, the size scaling $-0.064\,\%$ on both (26 of 30 and 41 of 49 better), and summed over dev30 its comfort term is unchanged: the gain is energy. Both are below the reseed resolution of a score; we keep them because the runs are paired and one-sided, and the size scaling improves the score under three occupancy seeds on the benchmark (-0.11, -0.09, $-0.05\,\%$). With occupancy

reseeded (12345, 777) the late start and drift clip move dev30 by -0.41 and $-0.38\,\%$, better on 16 scenarios, identical on 14, worse on none. On the shipped tree CtrlX scores 0.0069712 on dev30 ($-3.2\,\%$) and 0.004840 / 0.007353 on the holdouts ($-3.9\,\%$ pooled); on the leaderboard 0.01002 against 0.01032, $-2.9\,\%$.

The anchor. On the seven-day episode the public forum compares on (Appendix A) CtrlW scores -93.12, 11.2 units behind the best public physically consistent controller (-81.9 [22]). Neither information ($+2.60$ units for perfect per-zone foresight) nor the accounting seams close that, and our own 8 h receding-horizon LP on a 79-node twin moves the wrong way (-100.20; -94.15 with its defects fixed) and never discovers the release. The 12.8 units are a *sequencing* decision: released, the same controller reaches -80.31, past the best public number; -86.68 with the frost limit we ship, and CtrlX -84.63 with the same limit.

Ablations, reseeds, and what loses. Table 3 (appendix) adds one change at a time up to CtrlW5. The horizon sweep is non-monotone in the score—hour-06 comfort falls monotonically while the score does not (0.0075040, 0.0074263, 0.0074906, 0.0075614)—because readiness saturates once every zone is in band and each extra hour buys gas at night. Reseeding occupancy (17321, 12345, 777) moves CtrlW's dev30 score over 0.0073694–0.0073743 and CtrlW5-278's over 0.0071980–0.0072111 ($+0.18\,\%$ at most), while CtrlW5-278's gain over CtrlW reads -2.33, -2.33 and $-2.21\,\%$. What loses: holding the schedule's own setpoint (1.53 anchor units); a warmer 285 K night floor (6.9 units); the receding-horizon LP; and, for CtrlX, every uniform way of buying back the arrival hour—comfort is $1.0\,\%$ of its dev30 bill and $0.8\,\%$ is the hour after the mode change, yet a longer lead trades worse than one for one with gas, and the best of three leads per scenario, chosen with hindsight, is worth $0.02\,\%$ (Appendix C).

6 What information is worth

The observation space exposes one occupant count for the whole building, no per-zone occupancy and no weather forecast, while comfort is charged per zone. We hand the same controller each missing channel in turn (Table 6, Appendix D) and never submit any of them, so the deltas measure what sensing would be worth, and none of them is an entry. The oracles are transparent: dropping the payload reproduces an ordinary run to the last digit, and the forecast matches the sensor h hours later to 1.4×10^{-5} K.

Zonal occupancy pays $2.9\,\%$. Dropping each zone to standby 15 min after its own occupants leave, rather than waiting for the whole building to empty, improves dev30 from 0.0073694 to 0.0071585 and the corrected tree from 0.0071449 to 0.0069403 (both $-2.86\,\%$); on the ten-scenario subset it is $-2.30\,\%$, better on 10 of 10, recovering 23.9 units of energy plus carbon for 0.16 of comfort. Raising the release floor from 282 to 289 K moves the subset score by $0.09\,\%$: what matters is *which* zones are empty, and hardly how cold they are kept.

A perfect forecast is worth nothing on top of a clock rule. A fixed 2 K pre-dawn lift, gated only on the clock, the comfort-day predicate and the zone sitting on its standby floor, scores 0.0073463 on dev30 ($-0.31\,\%$) and is a wash on the corrected tree (0.0071582 against 0.0071449). Driving that lift and the preheat horizon from a perfect 12 h forecast scores 0.0073572: $0.15\,\%$ *worse* than the constant lift on dev30 and $0.004\,\%$ worse on the subset (the forecast-driven lift alone is $0.01\,\%$ better), so the information is worth zero to within our resolution.

The schedule is worth more than twice the sensor, and the difference is the lag. Per-zone foreknowledge of the episode's own occupancy, with a 5 h preheat lead, scores 0.0112001 on the subset ($-6.04\,\%$, 10/10) and -90.52 on the anchor ($+2.60$ units, $+2.79\,\%$); the same information five minutes late, as a sensor gives it, loses 3.7 points.

7 Benchmark integrity

The reward is computed from a simulator whose cost accounting is not closed under the policy's action space. Two seams are public and both have a proposed fix: the hot-water tank is billed from its setpoint rather than from its burner (B1, [22]), and the return air is mixed by floor area rather than by airflow (B2, [23]). Neither fix is merged and the leaderboard runs the unfixed tree, so what

each seam is worth is a measurement, taken by an instrument that leaves every per-step reward bit-identical (Appendix E).

B1 rewards an exploit. A probe applying the published setpoint recipe on top of CtrlW delivers the same $26.72\,\mathrm{MWh_{th}}$ of reheat while $11.1\,\mathrm{MWh}$ of gas stops being billed, comfort does not move, and the anchor improves by $+36.93$ units over CtrlW ($+39.7\,\%$), 3.3 times the gap to the best public score there. It is pure accounting: on the corrected tree the same probe is 0.59 units *worse*. Across dev30, gated to the base controller's own heating periods, the recipe takes CtrlW5-278 from 0.0071980 to 0.0058864 ($-18.2\,\%$; $-36\,\%$ on Fairbanks, 0 in Singapore) and costs $0.46\,\%$ corrected: a fifth of an honest score is free to whoever cycles the tank. A learned policy finds the seam untold: $39\,\%$ of a residual-PPO entry's gain was unbilled tank heat [4].

B2 hardly matters to a controller that moves air well. The manufactured heat is 0.7–$1.0\,\%$ of the coil heat the three well-scoring controllers buy, and the artifact is worth *minus* a fraction of a unit (-0.34, -0.42, -0.86; Table 7): harvesting it is priced out by the fan and by comfort. All three controllers improve when the accounting is fixed, more so on a mild scenario ($10.4\,\%$ on a full-length Atlanta episode), where the B1 probe reverses: its worth is scenario-dependent and the anchor figure is a ceiling. The third public channel is a leak: the occupancy generator runs on the kit's one default seed, which the observation does not expose, so replaying it would yield the per-zone schedule, whose value is the foresight number of §6 ($2.79\,\%$). Final verification reseeds, so we never use it.

B1 also overcharges an honest controller. When the base controller idles the loop after two hours without demand, the shipped tank falls to 310 K unbilled, and the first call of the next morning is billed for the recharge: in a mild seven-day week CtrlX is billed 1.055 units of gas per unit of heat its coils draw (1.017 in a cold one; 1.099 before the off-peak tank floor), and its shipped dev30 score is $1.9\,\%$ above its corrected one. The cleanest measurement is a change that does nothing physically. The burner raises the tank by at most $2.5\,\mathrm{K}$ per step in every scenario, so a setpoint capped at the observed tank temperature plus $2.5\,\mathrm{K}$ commands the same trajectory: on the corrected tree all ten benchmark scores are identical to the last digit, and on the shipped tree the capped controller scores $0.55\,\%$ better. Holding the tank setpoint through the idle branch and at its 30-hour peak, under that cap, removes the overcharge without ever billing less than the coils draw (1.035 and 1.004): the shipped score then meets the corrected one (Table 1) and the leaderboard entry moves from 0.01002 to 0.00982 (58217787). We report that variant apart, because it is a response to the accounting: it costs $0.05\,\%$ on the corrected tree and is worth nothing once the fixes land. A corollary for reading the public board: the late start of §4.5 is worth $0.33\,\%$ on the corrected development set and nothing on the leaderboard ($0.01007 \to 0.01008$), because the unfixed billing charges the later morning call. We attribute nothing to other teams beyond their published numbers: the audit bounds what these seams can be worth to a controller that respects comfort; it says nothing about what any particular team did.

8 Limitations and conclusion

The release relies on an empty building: with night occupancy, or a scenario colder than the coldest we ran, the unconstrained version buys at most 6 units at unbounded risk, which is why we ship a 278 K floor and report its price; it is also inert on the warm stations, whose air side already sits at its action floors (Appendix G). *The anchor overstates the effect*: the release gains $13.8\,\%$ over seven days but $2.4\,\%$ over the 30-day episode of the same building. *CtrlX leans on the benchmark's occupancy model*: nobody arrives on a public holiday and every zone holds the same headcount; with unequal headcounts the weight becomes people per square metre, which needs the sensor of §6. *The ceiling is the base model*: the largest CtrlX mechanism is worth $1.0\,\%$, and what is left of comfort is the arrival hour, which a one-node inversion cannot buy back.

Once a strong public controller is in hand, what is left to fix is the *schedule of the plant* and the *allocation of its margins*: stop heating at night and recover in one burst at the coil's cap, start late where the fabric is still charged, and insure comfort where it is cheap—0.0071449 $\to$ 0.0070409 $\to$ 0.0068387 on the corrected dev30, and better on 49 of 49 held-out scenarios.

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A The seven-day anchor

Table 2: The seven-day anchor the public forum compares on (`building_train_1/fairbanks_1`, 2 016 steps, mean outside $-26.3\,^{\circ}\mathrm{C}$, one seed), in reward units (higher is better, and so is a positive Δ). Public rows are other teams' published numbers on the same episode, not re-run by us, and we assume the default occupancy seed; *foresight* is analysis-only. CtrlMPC is our receding-horizon LP on a 79-node twin, v1 as first written and v2 with its defects fixed.

Controller	anchor total	Δ vs CtrlW	note
CtrlW (ours)	-93.12	—	readiness + weights + guards
CtrlW4 (ours)	-93.13	-0.004	pre-dawn lift, inert here
CtrlMPC v1 (ours)	-100.20	-7.08	8 h LP on the 79-node twin
CtrlMPC2 (ours)	-94.15	-1.02	LP defects fixed, no release found
CtrlW5-278	-86.68	$+6.44$	rooms $\geq 4.66\,^{\circ}\mathrm{C}$
CtrlW5-262 (no frost floor)	$\mathbf{-80.31}$	$\mathbf{+12.82}$	rooms to $-8.58\,^{\circ}\mathrm{C}$
CtrlX (shipped)	-84.63	$+8.49$	278 K floor
CtrlX + tank hold	-82.99	$+10.13$	278 K floor, §7
perfect per-zone foresight	-90.52	$+2.60$	analysis only
physical public controller [22]	-81.9	$+11.2$	published
twin + LP public planner [1]	-85.4	$+7.7$	published
free-action LP bound [1]	-79.1	$+14.0$	published bound

The planner rows in detail: the 8 h receding-horizon LP *buys* night capacity margin instead of selling it (night bucket $-1.22 \rightarrow -9.33$, day plan $+2.41$). Fixing its defects—heat priced at its physical cost, the coil's cold side at the zone rather than the mixed-air temperature, a plan margin and a night ceiling—recovers $+6.05$ of the 7.08 units and leaves it 1.0 unit behind CtrlW; allowed to float the night it still does not discover the release (-98.53). A twin-plus-LP public planner gets -85.4 and a free-action LP bound on the linearised building -79.1 [1]; the released controller is 1.6 units better than the best public number we know of and worse than that bound, which is an optimum and not an entry. B1 moves the best public controller by 0.3 units ($-81.9 \rightarrow -81.6$, as its author reports), so the gap was never an accounting one.

B Ablations

Table 3: Ablations against the public base controller on dev30 (shipped), one change at a time for a–d and then cumulatively, with the comfort component ($\times 10^{-4}$ per step) and the better/identical/worse count over the 30 scenarios (against the base for a–d, against CtrlW for the last four rows). h is the preheat horizon. CtrlW is a–d together; (a) and (d) are additive to within 1.3×10^{-6}.

Variant	score	Δ	comfort	better / identical / worse
base (Wu)	0.0075040	—	2.14	—
(a) occupancy release	0.0074459	-5.8×10^{-5}	2.15	23 / 7 / 0
(b) reward-consistent fuel weights	0.0075035	-5×10^{-7}	2.13	16 / 10 / 4
(c) holiday guard	0.0075040	0	2.14	0 / 30 / 0
(d) readiness $h = 2\,\mathrm{h}$	0.0074263	-7.8×10^{-5}	0.55	23 / 0 / 7
(d) readiness $h = 3\,\mathrm{h}$	0.0074906	-1.3×10^{-5}	0.31	16 / 0 / 14
(d) readiness $h = 4\,\mathrm{h}$	0.0075614	$+5.7 \times 10^{-5}$	0.30	10 / 0 / 20
CtrlW (a–d, $h = 2\,\mathrm{h}$)	0.0073694	$-1.8\,\%$	0.55	30 / 0 / 0
CtrlW2 (per-zone horizon)	0.0073576	$-0.16\,\%$	0.59	12 / 12 / 6
CtrlW4 (+2 K pre-dawn lift)	0.0073445	$-0.34\,\%$	0.87	8 / 12 / 10
CtrlW5-278 (+release, 278 K)	**0.0071980**	$-2.3\,\%$	1.06	18 / 12 / 0
CtrlW5 (+release, 262 K)	0.0071338	$-3.2\,\%$	1.41	18 / 12 / 0

C CtrlX: mechanisms, ablation and refutations

Protocol. The development benchmark is ten scenarios of 2 016 steps on the corrected tree with occupancy seed 12345: five shipped buildings and five generated ones, two per weather station. A run is kept only if the benchmark score falls, no room goes below $+4\,^{\circ}\mathrm{C}$, the module still imports

numpy and the standard library only and names no scenario, and the policy costs under 5 ms per step (it costs 1.6). CtrlW5-278 scores 0.0090209 there and CtrlX 0.0086829 ($-3.75\,\%$) after 52 logged runs and a longer list of probes; everything kept was then run paired on the 30-day sets on one machine. Table 4 lists the thirteen mechanisms and Table 5 the stages.

Table 4: The thirteen CtrlX mechanisms on the corrected tree. *7 d* is the change of the benchmark score when the mechanism was added. *30 d* is, for 1–11, the change of the `dev30` score when that mechanism alone is switched off in the controller that has all eleven (positive: the mechanism helps), with the number of scenarios that get worse / better without it; for 12 and 13, the paired change when it is added, with better / identical / worse. The ablation ran with the gate of mechanism 10 at 4 K; 6 K is a further $-0.07\,\%$ (10 / 19 / 1). Identical means within $0.005\,\%$.

#	mechanism	7 d, added	30 d
1	release when nobody has arrived $\lceil 330/V \rceil$ steps into a comfort day	$-0.94\,\%$	$+1.00\,\%$ (20 / 5)
2	load-driven hot-water reset, $(3 w_e \bar{c}/(w_g\, UA))^{1/4}$ above the supply air	$-0.37\,\%$	$+0.22\,\%$ (16 / 2)
3	no early pre-cooling of the upper bound	$-0.07\,\%$	$+0.09\,\%$ (8 / 5)
4	setpoint plus margin for the last 30 min before the mode change	$-0.21\,\%$	$+0.53\,\%$ (18 / 0)
5	model gain $0.75 \to 0.80$, and 0.75 while the hot-water pump saturates	$-0.6\,\%$	$+0.87\,\%$ (30 / 0)
6	comfort margin $0.12 \to 0.08$ K	$-0.33\,\%$	$+0.20\,\%$ (25 / 5)
7	economizer gate: early pre-cooling only when outside air can do it	$-0.01\,\%$	$+0.04\,\%$ (11 / 5)
8	night lift 2 K after 24 h with nobody in the building, else 1 K	$-0.18\,\%$	$+0.05\,\%$ (4 / 0)
9	per-zone margin earned by staying in band (up to 0.04 K)	$-0.26\,\%$	$+0.27\,\%$ (29 / 1)
10	late start for air-handler groups within 6 K of the setpoint	$-0.15\,\%$	$+0.21\,\%$ (15 / 0)
11	free-drift clip $-5 \to -8$ K per step	$-0.15\,\%$	$+0.12\,\%$ (4 / 4)
12	330 K hot-water floor only around the morning, 310 K otherwise	$-0.17\,\%$	$-0.033\,\%$ (15 / 14 / 1)
13	margin and arrival lead $\times(\bar{A}/A_v)^{1/2}$	$-0.11\,\%$	$-0.064\,\%$ (26 / 2 / 2)

Table 5: CtrlX by stage on the corrected tree, paired 30-day runs: `dev30` and the 49 holdouts (`gen` and `ext` pooled). Each holdout stage is better than CtrlW5-278 on all 49 scenarios; the last two steps are $-0.036\,\%$ (29 better / 13 identical / 7 worse) and $-0.064\,\%$ (41 / 1 / 7) there.

Stage	dev30 corrected	holdouts corrected
CtrlW5-278	0.0070409	0.0057936
+ mechanisms 1–9	0.0068733	0.0056222
+ 10, 11 (gate 6 K)	0.0068454	0.0055878
+ 12	0.0068431	0.0055858
+ 13 (CtrlX)	**0.0068387**	**0.0055822**

Why a gap gate (mechanism 10). Charging the fabric by Q while the air sits at T_{lo} instead of T_{hi} saves gas $Q\,[m(T_{\mathrm{hi}}) - m(T_{\mathrm{lo}})]$, with m the multiplier of Eq. (2); starting a group's plant Δt earlier to do so ventilates each of its zones at the floor flow, $0.3\,\dot{m} c_p\,(T_{\mathrm{zone}} - T_{\mathrm{out}})\,\Delta t$. Both scale with $T_{\mathrm{zone}} - T_{\mathrm{out}}$, so with $Q = C_f\, g$ for a gap g and a fabric capacity C_f per zone the early charge pays when $g > g^\star = \dot{m} c_p\,\Delta t \,/\, \big(C_f\,[1/(T_{\mathrm{dis}} - T_{\mathrm{hi}}) - 1/(T_{\mathrm{dis}} - T_{\mathrm{lo}})]\big)$. With the floor flow ($\approx 76$ W/K per zone), $\Delta t = 2$ h, $T_{\mathrm{hi}} = 294$ K, $T_{\mathrm{lo}} = 285$ K and C_f between the slab alone (3.8 MJ/K per 55 m^2) and slab plus partitions (≈ 10 MJ/K), $g^\star$ is 2–5 K. Measured, the gate is flat between 4 and 6 K and worse at 7–8 K on the one benchmark week whose groups straddle it. Leads must also be uniform inside a group: three staggered per-zone designs lost 1–7 %, because the plant then runs for one zone while the others are ventilated.

A finite-horizon credit (mechanism 12). Every episode starts with the hot-water tank at 340 K. On a traced mild week the off-peak floor saves 29 kWh of gas: about 8 kWh of standing loss and 23 kWh of heat that the controller without it still holds in the tank when the episode ends (324 K against 312 K), to within a kilowatt-hour. Three quarters of the seven-day gain is thus an initial condition converted into score, which any short-episode benchmark with storage rewards; on 30-day episodes the mechanism is worth $0.03\,\%$, and we report it as that.

The arrival hour. Comfort is $1.03\,\%$ of CtrlX's `dev30` bill and $0.81\,\%$ is the hour after the mode change (up to $4.7\,\%$ of a single scenario). The penalty is paid in the first two or three steps by the two to five people already in: on one traced morning five people are present, three of them in zones 0.3–0.7 K short, and that one step costs more than the plant's whole last half hour before the mode

change. The zones are not capacity-limited (valves at 0.35–0.45): the air jumps on the first step of the lead, sags as it relaxes to the fabric, and creeps 0.15 K per step. A uniform lead of 0.75 or 1 h instead of 0.5 h is $+0.13\,\%$ and $+0.43\,\%$ on `dev30` ($-0.18\,\%$ of comfort for $+0.31\,\%$ of energy; $-0.40\,\%$ for $+0.83\,\%$), and the best of the three per scenario, chosen with hindsight, is $-0.021\,\%$; a per-zone offset learned from the previous day's shortfall is flat to worse; a request beyond the clamped step is $+0.4$ to $+53\,\%$, with comfort lost in every later hour. Only the reallocation of mechanism 13 moved comfort and energy together.

Other refutations. Duty-cycling the frost floor (four designs, $+25$ to $+300\,\%$ of a $-37\,^{\circ}$C Saturday, and rooms below the floor); a per-zone model identified online by recursive least squares on the base model's regressors plus lags (its hindsight ARX fit predicts the next step three to five times better than the base model, the causal fit no better on two of four scenarios), which, inside the one-step search, tracks the setpoint from the first step of the lead and loses 1.5–$1.7\,\%$ in mild weeks; a late start on the cooling side ($+0.5$ to $+4.9\,\%$: the last pre-arrival hour of cooling pays); scaling the night lift by zone size ($+0.04\,\%$); and two CMA-ES campaigns over 14 knobs, whose optima are flat or worse on the benchmark.

D The information ladder

Table 6: The information ladder behind §6. Scores are the mean over scenarios of $|\frac{1}{T}\sum_t r_t|$ (lower is better): *subset* is ten scenarios at full episode length (its baseline is 0.0119199 because it overrepresents the coldest station, so only deltas compare across rows), dev30 the 30 shipped scenarios, *anchor* the seven-day comparison episode. Rows 3–7 read channels the competition does not expose and are analysis-only; row 2 needs no channel and is a legal controller. [†]the 1 K morning relax is a negative result (better on 5 of 10, comfort regret $5.33 \to 39.64$), included because it measures the first-arrival asymmetry directly.

#	what the controller knows	subset	Δ	dev30	anchor
1	nothing (baseline)	0.0119199	—	0.0073694	−93.12
2	+ fixed 2 K pre-dawn lift	0.0118597	−0.50 %	0.0073463	not run
3	zonal occupancy, 5 min late	0.0116456	−2.30 %	0.0071585	not run
4	zonal occupancy, 289 K floor	0.0116352	−2.39 %	not run	not run
5[†]	zonal occupancy + 1 K morning relax	0.0118677	−0.44 %	not run	not run
6	perfect 12 h weather forecast	0.0118603	−0.50 %	0.0073572	not run
7	perfect per-zone foresight (5 h lead)	0.0112001	−6.04 %	not run	−90.52

E The accounting audit, in full

The measurements behind §7, on the seven-day anchor, one seed. Both quantities are valid only at the instant the reward is billed, so they are captured by wrapping the model calls rather than by re-reading sensors (every read of the boiler's supply-water sensor advances the tank), and a policy's own trajectory is not perturbed: three rollouts of one episode with full hooks, with only the reward wrapper and with nothing give bit-identical per-step rewards (-93.12290938248043 shipped, -92.78747206777916 corrected), and on the corrected tree the instrument reports $+0.00$ kW of return-air imbalance (largest residual 2.2×10^{-13} W), reproducing the published fix rather than approximating it. The *worth* column is shipped minus corrected, i.e. what the artifacts are worth to that trajectory (negative means the controller is *better* on the corrected accounting). *Free heat* is the B2 imbalance a policy realises, as a share of the coil heat it buys.

Table 7: Accounting audit on the seven-day anchor (`building_train_1/fairbanks_1`, 2 016 steps, one seed), shipped tree against the tree with both public fixes. The upper block is the controllers that score well; the probes are analysis-only.

Policy	shipped	corrected	worth	free heat	gas billed (MWh)
ours (CtrlW)	-93.123	-92.787	-0.34	182 kWh (0.7 %)	$27.19 \to 27.09$
Wu (base)	-95.003	-94.581	-0.42	183 kWh (0.7 %)	$27.33 \to 27.21$
per-zone PI (zt-best)	-105.173	-104.317	-0.86	303 kWh (1.0 %)	$30.05 \to 29.64$
organizer `pid_water`	-596.865	-763.390	$+166.53$	1724 kWh (1.7 %)	$95.76 \to 124.53$
B1 probe (sawtooth)	-56.189	-93.380	$+37.19$	182 kWh (0.7 %)	$16.06 \to 26.98$
B2 probe (linear ramp)	-243.341	-255.838	$+12.50$	2370 kWh (4.6 %)	$51.38 \to 53.95$
B2 probe (2 kPa)	-4707.708	$-10\,258.068$	$+5550.36$	40 810 kWh (184 %)	$22.37 \to 32.82$

F Tools: an exact fast simulator

Every claim in the paper compares controllers on the same trajectory, so these studies are only as good as the number of episodes they can afford, and a speed-up that flips a damper decision is a different experiment. The whole of `env.step` now runs as one C call that reproduces the simulator's order-dependent behaviours: twelve closed-loop episodes (three buildings, both trees, two occupancy seeds) and full 8 641-step traces are bit-identical in observations and rewards, at 2.1, 1.8 and $1.6\times$ on the 30-, 79- and 85-zone buildings; five pure-Python patches recover 1.6–$1.9\times$ where C is unavailable. The shipped SuperLU is already optimal for a single right-hand side, batching is worth 4.1–$4.9\times$, a correct GPU solver loses to one CPU core, and a 200-mode reduced model drifts 1.0–1.2 K yet reproduces a five-way policy ranking exactly—a shortlisting filter, not a simulator (Table 8, Appendix F).

One instrumented step costs 28.1 ms of CPU on the smallest building (thermal solve 13.7, rest of the HVAC step 6.0, reward and observation assembly 8.4, plus the policy's 1.8 outside `env.step`), against the ≈ 125 ms per step the 9-hour rule allows. The fused step covers zone means, thermostat decisions, the damper floor, supply-air temperature, boiler and hot-water demand, gains, the finite-difference solve with its Woodbury air side, the air-side fold-back, convection, the observation vector and the reward numbers, with the policy-facing half (≈ 0.4 ms) and the array API (≈ 0.2 ms) left in Python. Two simulator behaviours dictated the design (the boiler's sensor mutates the tank on every read, and zone thermal power is assigned rather than accumulated), and the kernel reproduces both; in the twelve gate episodes $\max|\Delta r_t| = 0.0$ and zone temperatures agree to 1.2×10^{-15} relative. In interleaved A/B timing, cost per step falls from 20.5 to 9.6 ms on the 30-zone building, $49.8 \to 27.8$ on the 79-zone one and $67.6 \to 42.0$ on the 85-zone one.

Table 8 lists what each acceleration buys and at what exactness. The measurements behind it: the fused step's twelve closed-loop gate cases plus full 8 641-step traces on both trees; the five Python patches (cached Woodbury response replacing a 19.5 ms triangular solve with a 2.7 ms matvec, memoised zone means, memoised schedule and timezone queries, hoisted per-zone reward metadata, a cached gather index) at 1.6–$1.9\times$ for +170 MB per worker, inert when the fused step owns the thermal solve; the solver study (the per-step system is symmetric positive definite with condition number $\approx 1.4 \times 10^3$ and a Jacobi contraction bound of 0.9989, so an unpreconditioned iteration needs about 2 500 iterations for 10^{-12}, while a direct factorization built once per episode costs 0.2–1.2 s); the batched sweep saturating at $M = 32$–64 because it becomes main-memory bound (42 GB/s measured), with $M = 16$ already capturing 90 % of the win and, in the $M = 4$ gate, 19 of 20 trajectories identical (in the twentieth a damper decision flips at step 2 564 on a 10^{-14} field difference, and the episode total ends 3.5×10^{-9} relative away); the GPU verdict (smoothed-aggregation multigrid with a fixed 16-iteration PCG, 9.0×10^{-14} to 1.03×10^{-13} relative per step, 2.15×10^{-13} worst case chained, 22.3–23.5 ms per environment-step, the deficit being the toolchain rather than the algorithm, a fused V-cycle projected at ≈ 1.2 ms still behind the CPU path, and SuperLU's own batched solve *degrading* from 7.3 to 8.7 ms per environment-step as M goes from 1 to 64); and the reduced model (projection in the energy inner product $u = C^{1/2}T$, a plain Galerkin projection of $A^{-1}C$ producing spurious eigenvalues outside the unit disc and leaving the physical range within twelve steps, 1.0–1.2 K maximum drift and a -0.2 to -0.3 K bias flat from $m = 100$ to $m = 400$ that refitting does not remove, closed-loop reward off by 2.6 % and 5.0 % while the five-way ranking is exact, thermal step $14\times$ cheaper, end to end only $1.19\times$).

Table 8: What each acceleration buys, at what exactness. *Exactness* is measured against the unmodified simulator: a per-step relative error plus whether the episode total and the policy's decisions are unchanged. Speed-ups are interleaved A/B ratios on one machine; absolute times move with load, every ratio reproduced across sessions and on a second CPU.

Path	mechanism	speed	exactness	use
fused C step	one C call per step	$2.1/1.8/1.6\times$	rewards/obs. bit-identical, 12/12	our runner
`--fast`	5 memoizations	1.6–$1.9\times$	$\leq 2.8 \times 10^{-13}$ K; totals equal	local sweeps
LL^T, one RHS	METIS Cholesky	$0.9\times$	3.7×10^{-14}	keep SuperLU
LL^T, batched	M right-hand sides	4.1–$4.9\times$ per env-step	3.7×10^{-14}/solve; 19/20 equal	RL rollouts
GPU MG-PCG	16-iteration PCG	$0.3\times$ of 1 core	1×10^{-13}/step, 2.2×10^{-13} chained	no-go
ROM, $m = 200$	energy-norm projection	$14\times$ solve, $1.19\times$ e2e	1.0–1.2 K; ranking exact	shortlisting

G The bill, decomposed

The instrumented decomposition behind §3 (Figure 2), per full-length episode, in reward units (positive = loss). The outside-air column is the same step re-evaluated with the mixing box recirculating everything, i.e. the exact marginal effect of the outside-air fraction, with no assumption about coil saturation. Re-deriving cost and carbon from the logged powers reproduces the reward's own terms (4.9×10^{-10} relative for the worst combined term), and all six episodes reproduce the plain runner bit-for-bit.

Table 9: Energy-plus-carbon loss by mechanism, six climates (CtrlW, 8 641 steps each). Heating dominates the two Fairbanks buildings and chiller work the four warm ones; the outside-air share of the coil load is 50 % and 63 % in the two heating weeks, yet 100 % recirculation would cut coil duty by only 18.6 % and 9.3 %, because the coils are water-limited—the remainder is met by the fabric and appears as colder zones, one physical loss charged twice.

Scenario	$\bar{T}_{\mathrm{out}}$ (°C)	heating: OA	heating: envelope	boiler	cooling	fans	pump	total
Fairbanks, 39 zones	−21.4	69.65	306.69	5.08	0.00	8.62	2.51	392.55
Fairbanks, 85 zones	−8.2	9.29	90.55	2.12	0.00	4.06	0.66	106.69
Denver, 36 zones	24.2	0.00	0.16	1.56	32.70	2.97	0.00	37.39
Atlanta, 85 zones	20.8	0.01	0.39	0.91	19.44	3.83	0.00	24.58
Phoenix, 30 zones	34.4	0.00	0.00	0.07	78.06	4.87	0.00	83.00
Singapore, 54 zones	28.6	0.00	0.00	0.08	55.05	3.51	0.00	58.65

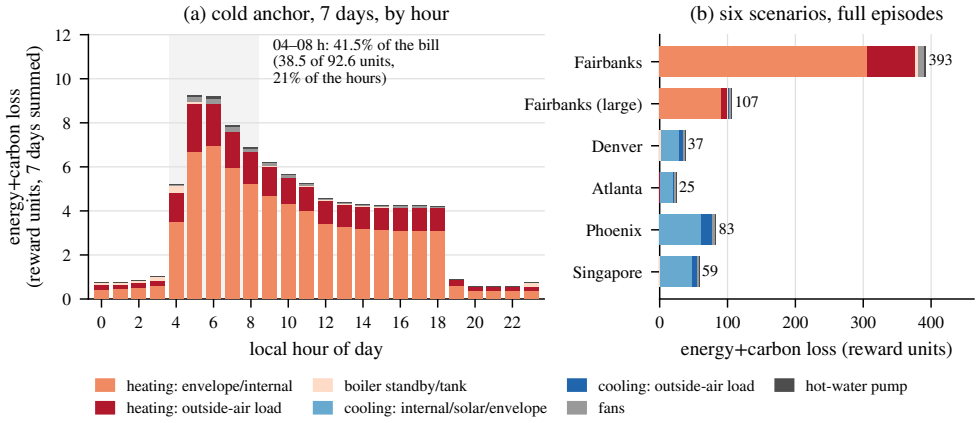


Figure 2: Reward anatomy for a controller whose comfort term is already 0.6 % of its loss. **(a)** The public anchor: energy-plus-carbon loss by local hour, stacked by mechanism—envelope reheat (light) plus the outside-air share of the same reheat (dark), 41.5 % of it in the pre-dawn 04–08 h block, 21 % of the hours. **(b)** The same decomposition for one full-length episode per climate; three scenarios hold 83 % of the 702.9-unit total (Appendix G). Each bar is a partition of the reward's own terms, reconciled below 1×10^{-9} relative.

The warm stations: nothing to release. The release is inert on the warm stations because both conditions it exploits are heating-only. On three seven-day warm anchors (CtrlW5-278, shipped)

the chiller is 93–95 % of the bill and the plant is already off when the building is empty (Phoenix: 0.5 % of chiller energy at night, 0.0 % at weekends). The coil is billed at a constant COP of 3.5 with no capacity limit, so there is no analogue of Eq. (2): only airflow matters, and both airflow floors bind—the supply-air setpoint sits at the 285 K bound of the action space on 100, 97.5 and 91.8 % of live AHU-steps (Phoenix, Singapore, Atlanta) and recirculation at the 0.70 device cap on 97–100 %. Raising the unoccupied ceiling is bit-identical (it is never reached); riding the band's top 0.12 K costs 5.0 units on Phoenix, because the margin is the tracking error and the reward prices an occupied kelvin at about 92 units a week against 0.62 of cooling; pre-cooling into cheap hours loses 3.9–6.9, because the electricity term is price-normalised (moving all of a week's cooling to its cheapest hour is worth 0.91 of 24.6 units); a night flush is inert where nights are warm and loses where it fires; and supply-air, reheat and static-pressure resets are outside the action space.

H Frost exposure

The measurements behind §4.4, on the six `dev30` Fairbanks scenarios at 8 641 steps, recorded by stepping the episode exactly as the evaluation runner does and reading only the zone-average temperatures (no monkeypatching, no extra reward call; the recorder reproduces the evaluation total to four decimals in every configuration where both exist). *h-any* is hours in which any room is below the threshold, *zone-h* room-hours.

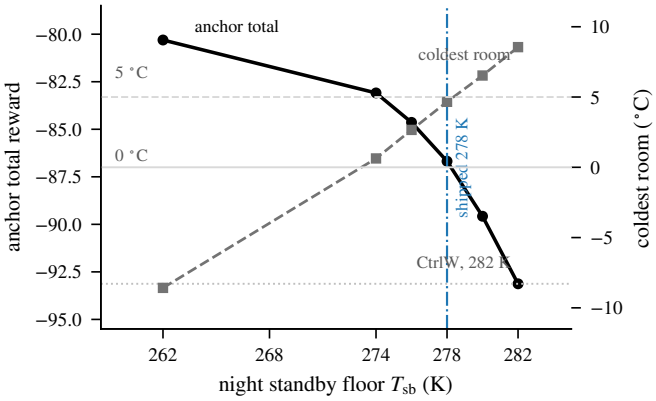


Figure 3: Price of a frost floor on the anchor (2 016 steps): anchor total reward (left) and the coldest room temperature the episode reaches (right) against the night standby floor, CtrlW5's other knobs fixed. On the anchor no room freezes from 274 K up; 278 K also keeps the three cold 30-day scenarios above +4.5 °C, and a ≥ 5 °C setpoint floor needs 280 K.

Table 10: Frost exposure and its price on the three cold `dev30` Fairbanks scenarios (`building_train_{1,2,8}/fairbanks_{1,2,8}`, 8 641 steps each); the other three never leave the band, so the floor is inert there. *h-any* is hours in which any room is below the threshold, *zone-h* room-hours, and *mean* the mean of $|\frac{1}{T}\sum_t r_t|$ over the three. Over all six Fairbanks scenarios the set scores are 0.015653 (CtrlW), 0.015063 (CtrlW5-278, −3.8 %) and 0.014733 (CtrlW5-262, −5.9 %).

Controller	coldest room	h-any < 5 °C	h-any < 0 °C	zone-h < 0 °C	mean	Δ vs CtrlW
CtrlW (282 K)	+8.53	0.0	0.0	0	0.027436	—
CtrlW5-278	+4.56	459.8	0.0	0	0.026384	−3.8 %
CtrlW5-262 (no frost floor)	−11.21	496.1	274.9	2681.8	0.025724	−6.2 %

I Reproducibility

Result directories contain one JSON per scenario with the score components, the hour-of-day and weekday/weekend decomposition and the metering-hook output; every entry we measured in Tables 1–10 is read from them, and Figures 1–3 are drawn from stored rollouts by `paper/figures/{multiplier,frost,anatomy}.py`, with no simulator calls. The base controller is ported from the public notebook [26] under Apache-2.0; with our flags off it reproduces our run of its author's notebook on dev30 to the last digit (0.0075040), which is the parity check we ran before adding anything.

Provenance. The shipped tree is the starter kit at commit `95cea5b` (2026-10-19, including the tracker fix of PR #1); the corrected tree adds PR #2 and PR #3 [22, 23], frozen as patches and applied by `sbx/make_sims.sh`. Generated buildings are the first 24 kit-generator seeds from 5000 that fall inside the shipped zone and air-handler ranges (5001–5025 for `gen_train`, 5026–5046 for `gen_hold`), materials seed 20261105 (`sbx/make_pool.py`); the external strips are described in Appendix J. Occupancy runs on the kit's seed 17321, and the reseeds use 12345 and 777; episodes are 8,640 scored steps plus a terminal one (result files report 8,641). The anchor CMA-ES (`cma` 4.5.0) searched CtrlW4's 16 knobs scaled to $[0, 1]$ at $\sigma_0 = 0.05$, population 12, seed 7, for 150 evaluations, starting from the best of 40 random points, with the default point injected in every population. The kit's other reference controllers are weaker than `pid_water` by its own shipped returns (fixed setpoint 0.0659, PI 0.0674, bang-bang 0.2252), a file that puts `pid_water` at 0.05414 against our re-run's 0.05498 (25 scenarios at 0.054363 and 5 re-runs at 0.058058), so read them as indicative. CtrlX is `policies/ctrl_x.py`, whose docstring and parameter comments state each mechanism with its evidence; every run of its development, kept or not, is logged in `docs/ctrlx_experiments.md`, and the result directories behind its numbers are under `results/ctrlx/`.

Commands. With `PY=starter-kit/.venv/bin/python` in the root of the repository (`https://github.com/MorS1337/sb26-stop-heating`; its README gives the setup):

```
$PY -m sbx.evalrun --policy policies.ctrl_w5:CtrlW5 --params '{"standby_minimum": 278.0}' \
    --set dev30 --steps 8641 --sim shipped --tag ctw5frost_dev30_shipped
$PY -m sbx.evalrun ... --sim corrected            # the corrected tree
$PY -m sbx.evalrun ... --set gen_hold ext_hold    # the 49 holdouts
$PY -m sbx.evalrun ... --cenv fused --fast        # the exact C step, same totals
$PY -m cenv.verify.gate trace --scenario building_train_8/fairbanks_8 --cenv fused --out t8_c
$PY -m kaggle.remote.remote_eval --policy ... --set dev30 --shards 4 --cenv fused --fast
$PY kaggle/build_notebook.py --policy policies/ctrl_w5.py:CtrlW5 \
    --param standby_minimum=278.0 --slug sbx-ctrl-w5f   # the CtrlW5-278 leaderboard entry
$PY -m sbx.arbench                                    # the CtrlX development benchmark
$PY -m sbx.evalrun --policy policies.ctrl_x:CtrlX --set dev30 --steps 8641 --sim corrected \
    --cenv fused --fast --tag ctrlx_dev30_corrected   # CtrlX; add --params for an ablation
$PY kaggle/build_notebook.py --policy policies/ctrl_x.py:CtrlX --slug sbx-ctrl-x   # 58217472
$PY kaggle/build_notebook.py --policy policies/ctrl_x.py:CtrlX --slug sbx-ctrl-x-hold \
    --param water_idle_hold=true --param water_peak_hold_hours=30.0 \
    --param water_rise_guard_K=2.5 --param water_offpeak_floor=0.0      # tank hold, 58217787
```

Machines and numerics. Local runs used an Apple M3 and a Ryzen 5 5600X (WSL2); the 30-scenario and holdout sweeps also ran on Kaggle CPU kernels, at most four at a time. The fused step is compiled with `-O3 -ffp-contract=off -fno-fast-math`—no FMA contraction, no reassociation—the flags under which it is bit-identical to the Python step on each machine. Across machines (arm64 with Accelerate against x86-64 with OpenBLAS) the same episode differs by up to 3.3×10^{-5} relative over 30 days—round-off amplified by the closed loop, e.g. -93.12290938 against -93.12336623 for the seven-day anchor—far below the printed precision of S and the reseed spread, but enough to turn "identical" into "better" or "worse"; every better/identical/worse count therefore compares runs from one machine.

J External weather

`ext` replays hourly observations from Meteostat [14] for calendar year 2023 at seventeen stations absent from the benchmark—Anchorage, Auckland, Bangkok, Beijing, Chicago, Dubai, Helsinki, Houston, Las Vegas, London, Miami, Montreal, Moscow, Reykjavik, San Francisco, Seattle and Tokyo—mapped to the benchmark's weather-file schema with no other scenario field modified. Meteostat data are distributed under CC BY 4.0; derived weather files are redistributed under the same terms with attribution to Meteostat and its data providers.